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$$\begin{aligned} & \lim_{x \rightarrow 2} \left(\frac{3x-2}{2x} \right)^{\frac{1}{x-2}} \\ &= \lim_{x \rightarrow 2} \left(\frac{x-2+2x}{2x} \right)^{\frac{1}{x-2}} \\ &= \lim_{x \rightarrow 2} \left(1 + \frac{x-2}{2x} \right)^{\frac{1}{x-2}} \\ &= \lim_{x \rightarrow 2} \left(\left(1 + \frac{x-2}{2x} \right)^{\frac{2x}{x-2} \cdot \frac{x-2}{2x}} \right)^{\frac{1}{x-2}} \\ &= e^{\lim_{x \rightarrow 2} \frac{\frac{x-2}{2x}}{\frac{x-2}{2x}}} \\ &= e^{\lim_{x \rightarrow 2} \frac{1}{2x}} \\ &= e^{\frac{1}{4}} \end{aligned}$$

References